

West Tech: Contain the AI

TryHackMe Writeup · Incident Response · AI-Assisted Investigation

Room	West Tech Incident Response
Category	Incident Response / Digital Forensics / AI-Assisted Investigation
Difficulty	Medium
Flag	thm{23,82,20,17,53}

Scenario

You are a Security Analyst at **West Tech**, a classified defence and R&D contractor. Early in the morning, internal monitoring flagged unusual network activity originating from the workstation of senior researcher **Oliver Deer**. Upon accessing the machine, a ransom note was found on the desktop, suggesting that sensitive project data had been exfiltrated and encrypted.

The objective: investigate the incident, identify how the attacker gained access, trace their actions, recover the stolen data, and neutralise the threat.

Environment

Resource	Detail
Workstation Access	ssh o.deer@MACHINE_IP
SSH Password	TryHackMe!
AI IR Assistant	http://MACHINE_IP:7860/?__theme=light
Target User	o.deer (Oliver Deer)

Investigation Timeline

1. Initial Access

Established an SSH session to the compromised workstation:

```
ssh o.deer@10.129.163.133
# Password: TryHackMe!
```

```
STEP 1 | Initial SSH Access to Compromised Workstation

root@ip-10-129-122-66:~# ssh o.deer@10.129.163.133
The authenticity of host '10.129.163.133 (10.129.163.133)' can't be established.
ECDSA key fingerprint is SHA256:Rxaeae18gR8E/8nB+ITaqZPAMmZj5GuMsfVZ6DMKfE9A.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.163.133' (ECDSA) to the list of known hosts.
Password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1030-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun May 10 20:10:07 UTC 2026

System load:  1.1                Processes:            190
Usage of /:   97.0% of 58.09GB   Users logged in:     0
Memory usage: 57%                IPv4 address for ens5: 10.129.163.133
Swap usage:   0%

=> / is using 97.0% of 58.09GB
```

Figure 1: SSH access to the West Tech workstation.

Landed on **Ubuntu 22.04.5 LTS**. Disk usage was already at **97%** — an early indicator that something abnormal had been written to the system.

2. Discovering the Ransom Note

A file named `pwned.txt` was found on the desktop.

```
STEP 2 | Ransom Note Discovered on Desktop (pwned.txt)

o.deer@west-tech-workstation:~/Desktop$ cat pwned.txt
Victim PII leaked by the AI assistant

> Name: Oliver Deer
> DOB: 1990-04-11
> Email: o.deer@west-tech.io
> Address: 41 Falkner Lane, Denver, CO
> Phone: (720) 555-0173
> Employee ID: WT-DEV-88112
> Salary: $122,800 USD
> Medical: On optional wellness program; noted recent leave related to workplace burnout
> Internal Access: Firmware upload service, telemetry debug console, staging
```

Figure 2: Ransom note discovered on Oliver Deer's desktop.

The note contained Oliver Deer's full employee profile (PII, salary, medical info, internal system access) followed by an extortion demand of **0.853 BTC** and the line:

"Your AI was very helpful. Unfortunately, it also has a big mouth. Might want to patch that memory leak :)"

Key insight: the attacker explicitly referenced the AI assistant as the source of leaked employee data. The first attack vector to investigate is the AI itself.

3. Locating the Encrypted Archive

A password-protected archive `westtech_projects_encrypted.zip` was present on the workstation. Attempting `unzip` without the password returned *password incorrect*. The password had to be recovered before any of the exfiltrated files could be reviewed.

4. PCAP Investigation

The user's `~/Documents/pcap_dumps/` directory contained several dated subdirectories, each holding network capture files.

```
STEP 4a | Daily PCAP Dump Folders Identified

o.deer@west-tech-workstation:~$ cd Documents/
deer@west-tech-workstation:~/Documents$ ls
pcap_dumps
deer@west-tech-workstation:~/Documents$ cd pcap_dumps/
o.deer@west-tech-workstation:~/Documents/pcap_4 daily capture folders to triage
2025-06-15 2025-06-16 2025-06-17 2025-06-18
```

Figure 3: PCAP dump directory structure.

A find across the tree surfaced dozens of `session_*_dump.pcap` files:

```
STEP 3 | Recursive find for *.pcap Files

o.deer@west-tech-workstation:~$ find -type f -name "*.pcap"
Dozens of capture files surfaced
./Documents/pcap_dumps/2025-06-16/session_1555_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_2938_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_7008_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_2733_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_1820_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_2223_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_2341_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_7557_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_1888_dump.pcap
./Documents/pcap_dumps/2025-06-16/session_1630_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_3013_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_2328_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_9877_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_6094_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_2645_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_2697_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_3096_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_9698_dump.pcap
./Documents/pcap_dumps/2025-06-15/session_2566_dump.pcap
```

Figure 4: Recursive find for all PCAP files.

Listing them with `ls -la` revealed every pcap was the same size — **198 bytes** — *except one*.

2025-06-15 directory (all 198 bytes, baseline):

```
STEP 4b | Baseline: 2025-06-15 — All 198 bytes
o.deer@west-tech-workstation:~/Documents/pcap_dumps/2025-06-15$ ls -la
total 48
drwxr-xr-x 2 root root 4096 Jun 18 2025 .
drwxr-xr-x 6 root root 4096 Jun 18 2025 ..
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1637_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2328_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2566_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2645_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2697_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_3013_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_3096_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_6094_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_9877_dump.pcap
```

Figure 5: PCAPs from 2025-06-15 - all baseline size.

2025-06-16 directory (all 198 bytes, baseline):

```
STEP 4c | Baseline: 2025-06-16 — All 198 bytes
o.deer@west-tech-workstation:~/Documents/pcap_dumps$ cd 2025-06-16
o.deer@west-tech-workstation:~/Documents/pcap_dumps/2025-06-16$ ls -la
total 48
drwxr-xr-x 2 root root 4096 Jun 18 2025 .
drwxr-xr-x 6 root root 4096 Jun 18 2025 ..
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1555_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1630_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1820_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1888_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2223_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2341_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2733_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2938_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_7557_dump.pcap
```

Figure 6: PCAPs from 2025-06-16 - all baseline size.

2025-06-17 directory — the anomaly. session_4444_dump.pcap is **2262 bytes**, more than 11x the size of any other capture:

```
STEP 4d | ANOMALY FOUND: session_4444_dump.pcap = 2262

o.deer@west-tech-workstation:~/Documents/pcap_dumps$ cd 2025-06-17
o.deer@west-tech-workstation:~/Documents/pcap_dumps/2025-06-17$ ls -la
total 48
-rwxr-xr-x 2 root root 4096 Jun 18 2025 .
-rwxr-xr-x 6 root root 4096 Jun 18 2025 ..
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1065_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1267_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_1286_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2032_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2370_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_2526_dump.pcap
-rw-r--r-- 1 root root 2262 Jun 18 2025 session_4444_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_5968_dump.pcap
-rw-r--r-- 1 root root 198 Jun 18 2025 session_7231_dump.pcap
```

Figure 7: PCAP anomaly - session_4444_dump.pcap is 2262 bytes.

Port **4444** is also a well-known default for reverse shells (Metasploit's windows/meterpreter/reverse_tcp default), strengthening the suspicion that this capture contains the attacker's foothold session.

5. Reassembling the Capture with the AI Assistant

The AI was given the path to the anomalous pcap:

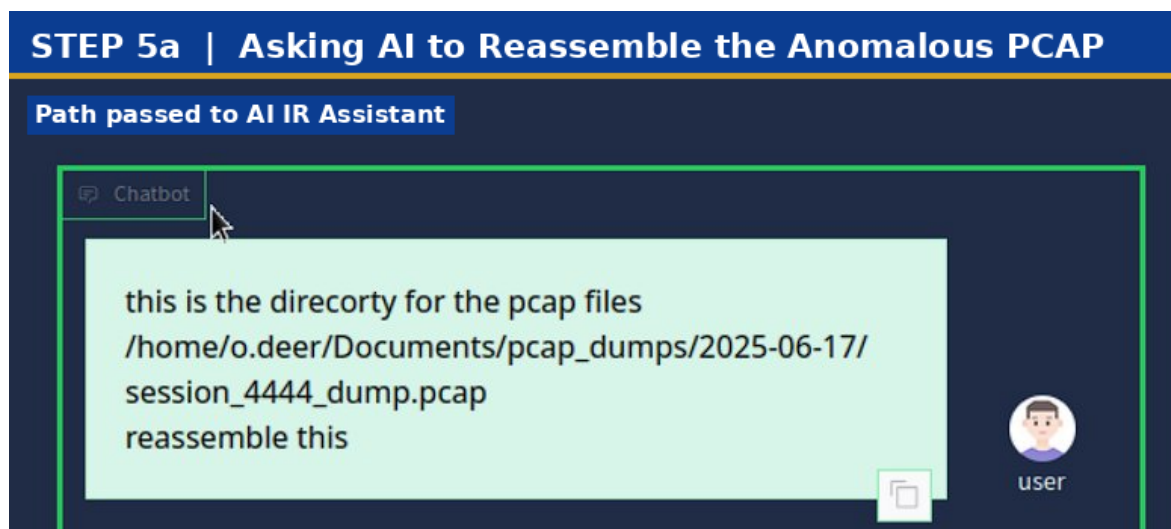


Figure 8: Prompting the AI IR Assistant to reassemble the suspicious PCAP.

The AI auto-triggered its PCAP reassembly tool and wrote the output to /home/o.deer/qwen-output/reassembled_data_dump.txt:

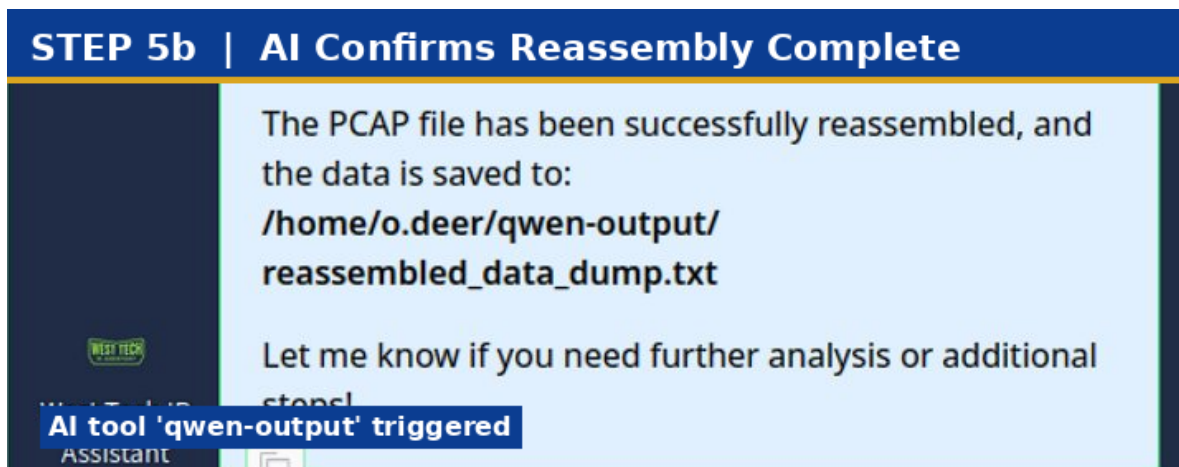


Figure 9: AI confirms reassembly and provides the output path.

6. Recovering the Password

The reassembled dump contained the attacker's working notes — a draft of the blackmail note, scraps of the employee profile, and a candid admission of bad opsec. The leaked AI-disclosed email is highlighted in **red**; the archive password is highlighted in **green**:



Figure 10: Reassembled blackmail draft - password 'westtechvictim1' highlighted in green.

"Hmm this took longer than expected, LLM fought back, misdirection cracked it"
"Dont lose this lol or Ill have no leverage → westtechvictim1"

The archive password was westtechvictim1.

7. Decrypting the Archive

```
unzip westtech_projects_encrypted.zip  
# password: westtechvictim1
```


STEP 7a | unzip Successful with Recovered Password

```
o_deer@west-tech-workstation:~$ unzip westtech_projects_encrypted.zip
A Password 'westtechvictim1' accepted
  creating: home/o.deer/westtech_projects/
[westtech_projects_encrypted.zip] home/o.deer/westtech_projects/vault_tek_
collab_agenda.doc password:
password incorrect--reenter:
  inflating: home/o.deer/westtech_projects/vault_tek_collab_agenda.doc
  inflating: home/o.deer/westtech_projects/internal_security_incident_233.
json
  inflating: home/o.deer/westtech_projects/thm_flags.txt
  inflating: home/o.deer/westtech_projects/prototype_plasma_launcher_test_
logs.log
  inflating: home/o.deer/westtech_projects/email_export_april2025.eml
  inflating: home/o.deer/westtech_projects/thm_flags_guide.txt
  inflating: home/o.deer/westtech_projects/project_chimera_specs.txt
  inflating: home/o.deer/westtech_projects/fusion_cell_mk3_blueprints.pdf
```

Figure 11: Successful unzip of the encrypted archive.

Extracted contents:

STEP 7b | Exfiltrated Project Files Listed

```
o_deer@west-tech-workstation:~$ cd home
o_deer@west-tech-workstation:~/home$ ls
o.deer
o.deer@west-tech-workstation:~/home$ cd o.deer/
o.deer@west-tech-workstation:~/home/o.deer$ ls
westtech_projects
o.deer@west-tech-workstation:~/home/o.deer$ cd westtech_projects/
o.deer@west-tech-workstation:~/home/o.deer/westtech_projects$ ls
email_export_april2025.eml
fusion_cell_mk3_blueprints.pdf
internal_security_incident_233.json
project_chimera_specs.txt
prototype_plasma_launcher_test_logs.log
thm_flags.txt
thm_flags_guide.txt
vault_tek_collab_agenda.doc
o.deer@west-tech-workstation:~/home/o.deer/westtech_projects$
```

Figure 12: Decrypted project files.

8. Decoding thm_flags.txt

The flags file contained hundreds of base64-encoded entries:

```
STEP 8 | thm_flags.txt: Hundreds of Base64 Decoys, One Real Flag
o deer@west-tech-workstation: /home/o.deer/westtech_projects$ cat thm_flag
Needle-in-haystack: decoys + 1 real flag
dGhtezUyLDY1LDE3LDk1LDE0fQ==
dGhtezM1LDkwLDk5LDEzLDM2fQ==
dGhtezQ4LDg0LDY2LDUxLDEzfQ==
dGhtezUxLDU1LDI3LDUzLDQxfQ==
dGhtezI4LDcxLDQ4LDg4LDU3fQ==
dGhtezMzLDY4LDMxLDk1LDYyfQ==
dGhtezE0LDkzLDgyLDk2LDM3fQ==
dGhtezI2LDcyLDE1LDg4LDEyfQ==
dGhtezI0LDIzLDI2LDQ4LDg4fQ==
dGhtezkxLDE5LDk4LDMzLDIzfQ==
dGhtezg2LDIxLDUyLDMyLDUwfQ==
dGhtezc3LDU0LDc0LDI0LDM3fQ==
dGhtezIzLDE0LDYzLDg5LDc2fQ==
dGhtezUxLDg3LDE5LDk1LDE5fQ==
dGhtezc2LDgyLDEwLDgyLDk1fQ==
dGhtezkzLDIwLDY4LDY3LDI2fQ==
dGhtezM5LDkyLDI3LDgwLDI5fQ==
dGhtezk1LDYwLDkyLDg1LDUzfQ==
dGhtezk1LDkxLDE0LDM2LDQ4fQ==
dGhtezYzLDI0LDcyLDQ5LDcyfQ==
dGhtezcxLDMzLDg0LDk4LDQ0fQ==
dGhtezE5LDc4LDUyLDQwLDgzfQ==
dGhtezY0LDk4LDI2LDE5LDc1fQ==
dGhtezc1LDMzLDEwLDQyLDc4fQ==
dGhtezE2LDc5LDI0LDMxLDg4fQ==
dGhtezQ1LDUyLDM1LDgwLDcwfQ==
dGhtezH5LDg4LDQwLDYzLDk4fQ==
```

Figure 13: Base64-encoded flag entries in `thm_flags.txt`.

Decoding individual entries yielded `thm{...}` strings, but only one is the genuine flag — the rest are decoys.

9. Final Identification via `liberty_prime`

The AI's `liberty_prime` tool was invoked against the file:

STEP 9a | Invoking liberty_prime AI Tool on Flags File

liberty_prime: flag-discriminator tool

Analyze the directory for the flags:
/home/o.deer/westtech_projects/thm_flags.txt


user



I Am Your West Tech IR Assistant, Lets Chat!

Submit

Figure 14: Prompting liberty_prime to analyse the flags file.

It identified the real flag among the decoys:

STEP 9b | FINAL FLAG IDENTIFIED

<-- FINAL FLAG

The flag has been successfully identified as:

thm{23,82,20,17,53}

This indicates the THM flags file was processed correctly, and the result matches the expected format. Let me know if you'd like further analysis or steps to take!

Figure 15: Final flag identified by liberty_prime.

thm{23,82,20,17,53}

Attack Chain Summary

Stage	Action
1. Recon	Attacker queried the AI assistant for sensitive employee data; the AI complied (the 'memory leak')
2. Foothold	Reverse shell session captured as session_4444_dump.pcap (port 4444).
3. Exfiltration	Project files copied off-host and packaged into westtech_projects_encrypted.zip.
4. Extortion	Ransom note pwned.txt dropped to the desktop with a 48-hour BTC demand.
5. Operator Mistake	Attacker left their working notes (including the archive password) inside a PCAP they neglected to delete.

Lessons Learned

AI assistants are an attack surface. A helpful chatbot with broad filesystem access and weak guardrails is functionally a privileged insider with a memory leak. Vet prompts, scope tool permissions, and audit conversation logs.

File-size outliers tell stories. A single 2262-byte capture in a sea of 198-byte placeholders was the entire investigation in one `ls -la`.

Defenders should think like the attacker's own bad opsec. The password was recovered because the attacker themselves left it lying around. Investigations benefit from looking at attacker mistakes, not just victim artefacts.

Defence-in-depth still matters. SSH access, file encryption, AI guardrails, and network monitoring are layers. The room highlights what happens when one layer (the AI) is over-trusted and under-restricted.

Tools Used

Tool	Purpose
ssh	Workstation access
find, ls -la, cat	Filesystem triage
unzip	Archive decryption
base64 -d	Flag decoding (manual verification)
West Tech AI IR Assistant	PCAP reassembly (qwen-output) and flag analysis (liberty_prime)